

Vitamins

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Vitamins

- Are organic compounds required in the diet in small amounts to perform specific biological functions for normal maintenance of optimum growth and health of the organism.
- Useful to correct deficiencies but taking higher doses of vitamins will not boost up the health

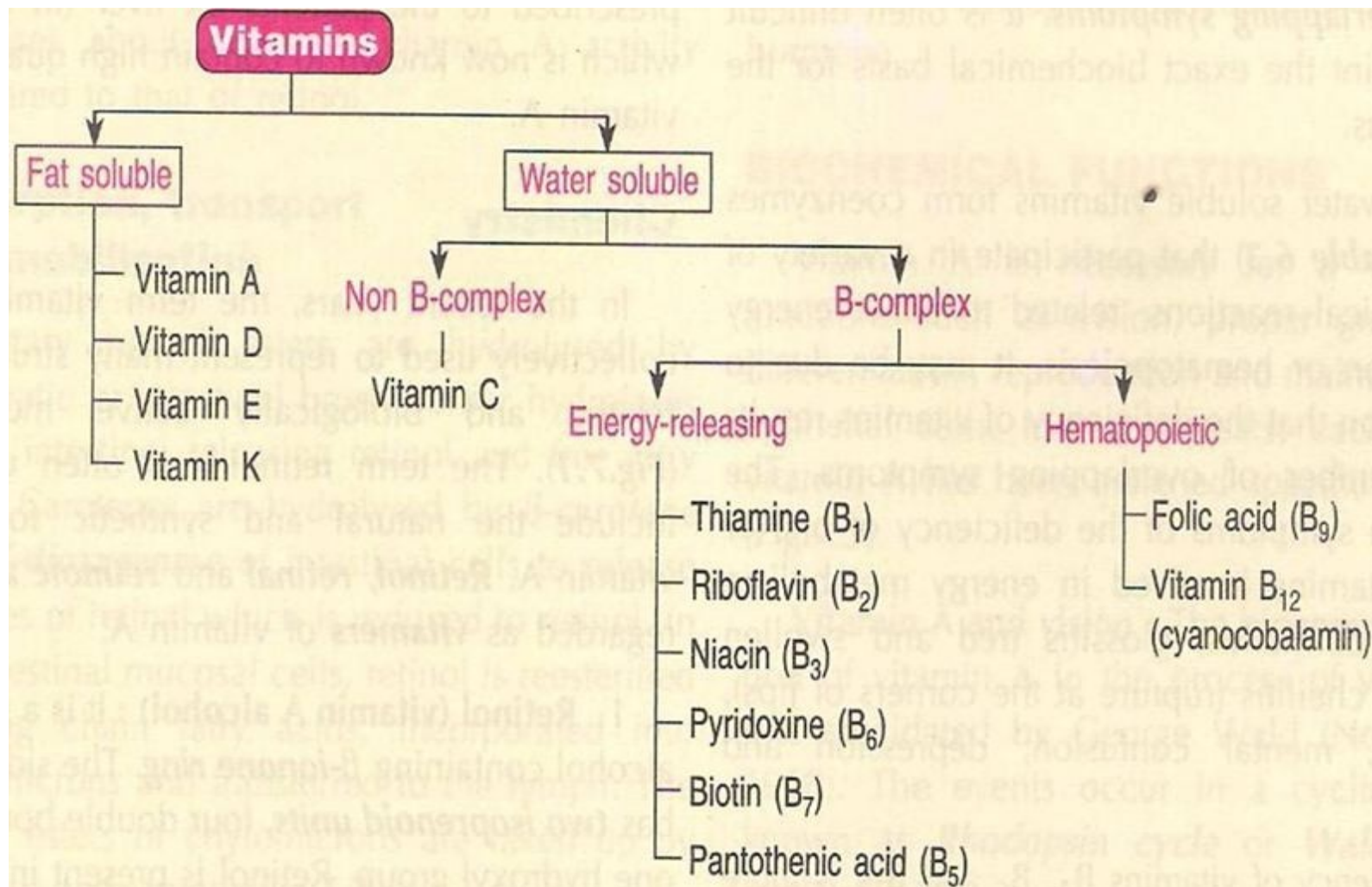
❖ The importance of vitamins

- ❑ Many vitamins function biochemically as precursors in the synthesis of coenzymes necessary in human metabolism.
- ❑ Therapeutic indications for diseases caused by certain vitamin deficiencies
- ❑ There are products indicated for prophylactic use as dietary supplements

❖ Deficiency could lead to serious diseased conditions

- – Beriberi; Scurvy; Rickets; Pellagra; Night blindness etc.

Classification of Vitamins



Lipid soluble vitamins

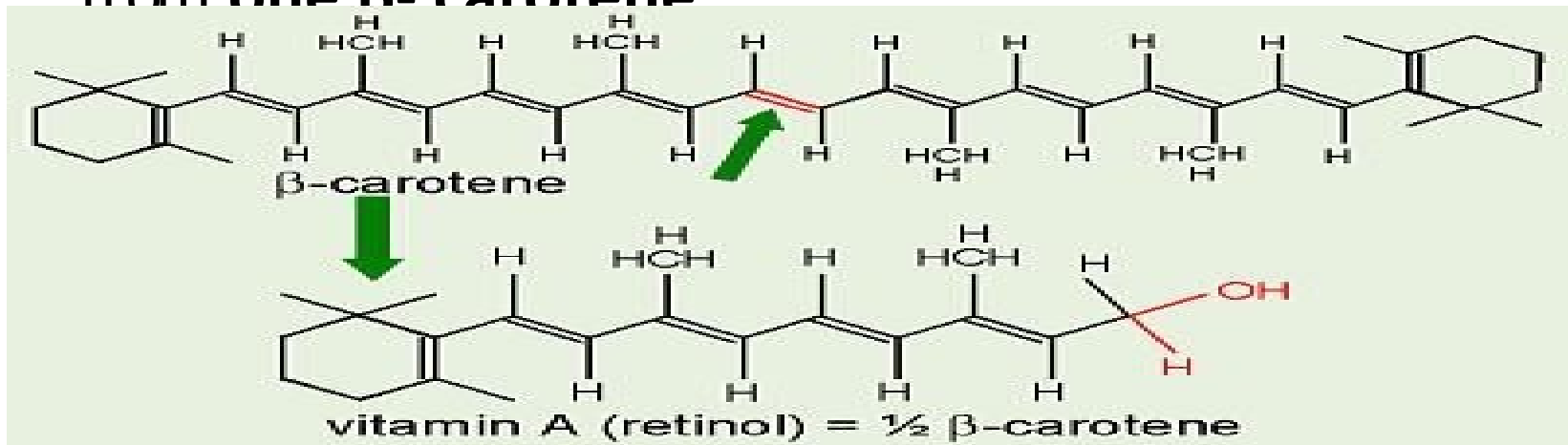
- They are usually associated with lipids of foods and are absorbed from the intestine with these dietary lipids
- They are stored in the liver & thus conserved by the organism.
- Single large doses may prevent deficiencies (If any)

Vitamin A

- Often called growth vitamin due to a deficiency of it causes a cessation of growth.
- Stored principally in the liver as **retinol palmitate**
- All compounds with vitamin A activity are referred as **Retinoids**
- They are polyisoprenoid compounds having a β -ionone or cyclohexenyl ring system

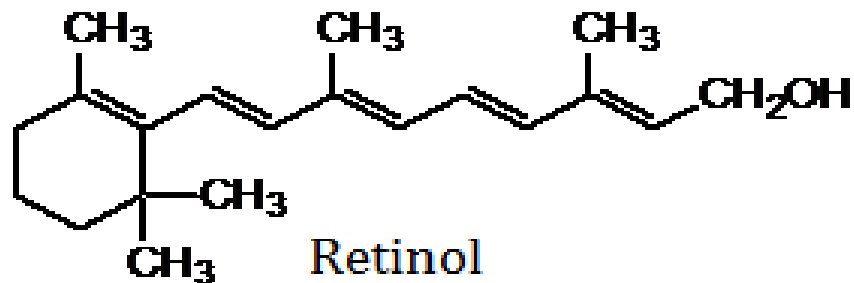
Vitamin A

- The **intestinal mucosa** is the main site of **β -carotene transformation to retinol** by an enzyme **dioxygenase** using molecular oxygen (found also in hepatic tissue) responsible for the formation of **two retinol molecules** from **one β -carotene**



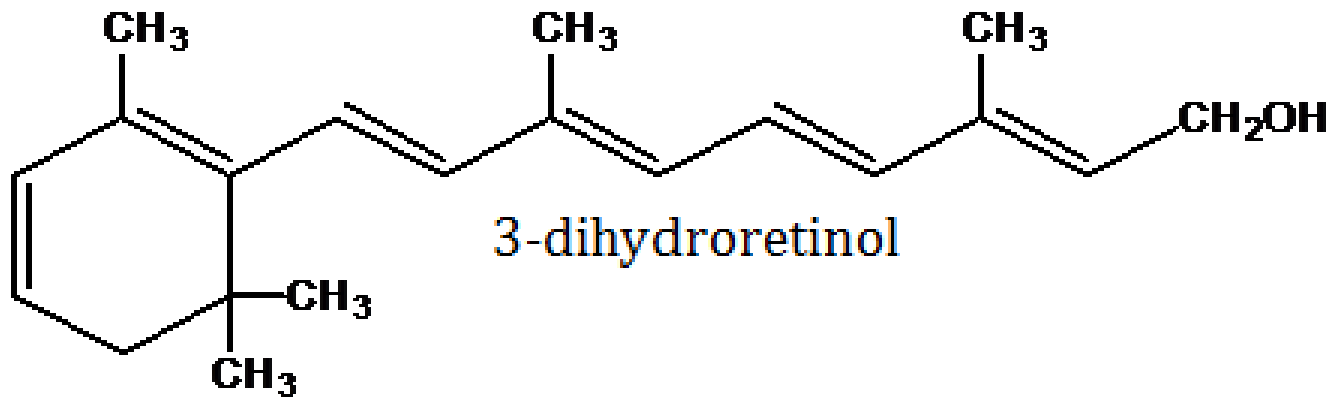
Vitamin A

- The retinol molecule undergoes **oxidation** to retinal by retinal reductase, which is reversible
- Retinoic acid promotes
 - The development of bone & soft tissues
 - Sperm production
 - But does not participate in the visual process
- The side chain contains alternate double bonds and hence, many isomers are possible

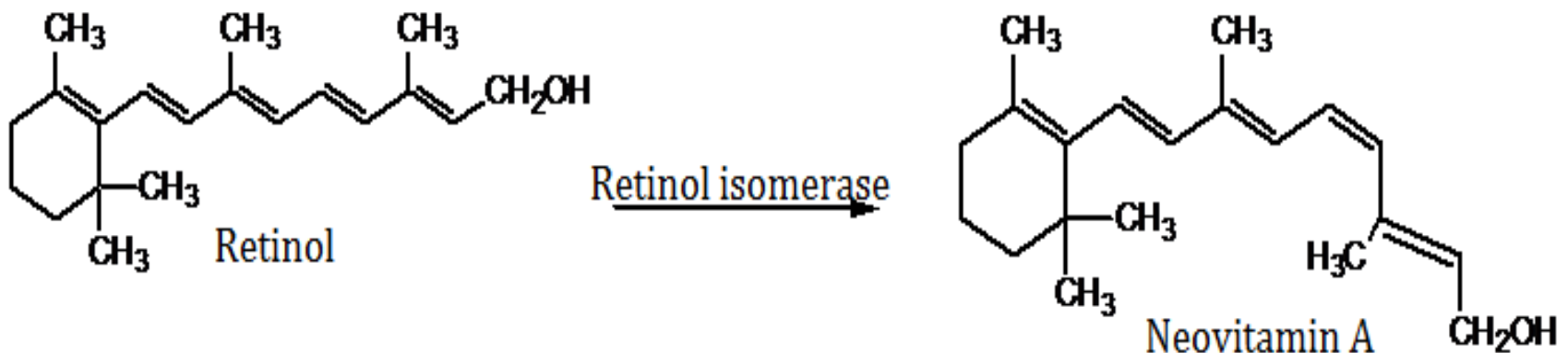


Vitamin A

- Vitamin A₂ (Dihydroretinol)
 - It exhibits chemical, physical & biological properties very similar to those of Vit. A₁
 - Liver of fresh water fish contains Vitamin A₂



- A cis- isomerism at the 11th position of vitamin A of the side chain produces a biologically important compound, Neovitamin A (11- mono cis retinol)
 - Retinol isomerase converts vitamin A to neovitamin A in the liver
 - Neovitamin A is the active isomer on retina photoreceptor cells, active in



Biological (Biochemical) role of vitamins A

- Required for development & maintenance of epithelial tissue (Growth & differentiation of tissues)
- Enhances the function of immune system
- Responsible for vision
- Necessary for reproductive system

Deficiency of vitamin A

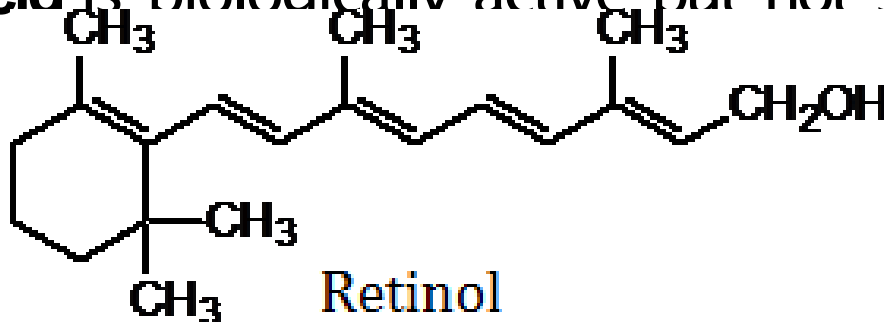
- In the case of vitamin A deficiency, the retinol- binding protein (RBP) level in blood falls
- The deficiency of vitamin A manifests, only if the diet is devoid of vitamin for a period of 2- 3 months, when the hepatic storage is depleted
 - Causes keratomalacia- softening of the cornea
- It is manifested chiefly by degeneration of the mucous membrane in the eye causing dry eye, a condition known as Xerophthalemia -ocular defects
- Development of night blindness (Nyctalopia) in the early stages of vitamin A deficiency, which leads to blindness
- Advanced deficiency causes dryness and scaliness of the skin

Massive overdose of Vitamin A

- It can cause
 - Rough skin,
 - Dry hair,
 - Enlarged liver,
 - Raised serum calcium & alkaline phosphate concentration

Structural activity relationship vitamin A

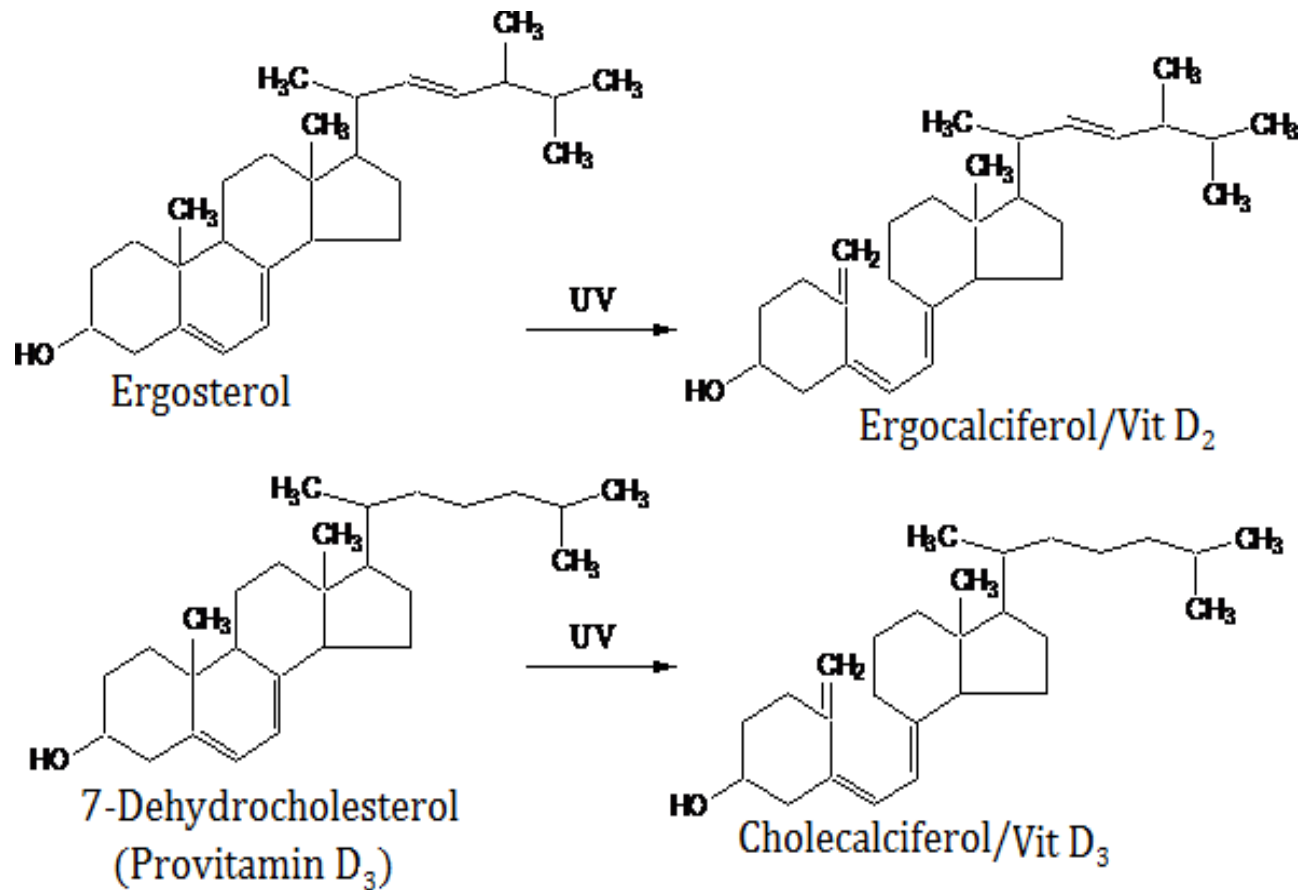
- The **conjugated double bonds** in Vitamin A are all essential for activity
- The β -ionone ring is essential for activity
 - Both in retinol & dihydroretinol
- The **ester or methyl ether** of Vit. A are equally active to Vit. A
- **Retinoic acid** is biologically active but not stored in the liver



Vitamin D

- Also called
 - Sunshine vitamin
 - Antirachitic factor
- It resembles sterol in structure and functions like a hormone
- **Vitamin D** is derived either from **7-dehydrocholesterol** or **ergosterol** by action of UV-radiation.
 - Found to be a mixture of isomers: **Lumisterol (Vitamin D₁)** &
 - **Calciferol (Vitamin D₂) (Ergocalciferol)**

Formation of vitamins D₂ & D₃

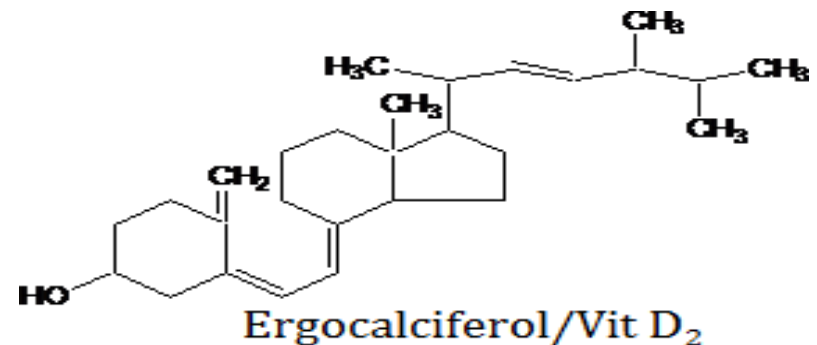
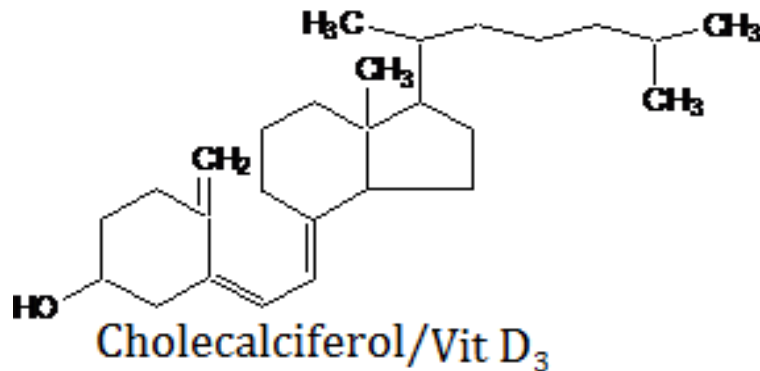


Uses of Vitamin D

- Vit. D & their active metabolites aid in
 - Facilitating absorption of Ca^{2+}
 - Re-absorption of PO_4^{2-} into the renal tubule
- Necessary for normal growth in children
 - Have direct effect on the osteoblast cells
- Important in treatment of hypoparathyroidism
- Deficiency leads to
 - Inadequate absorption of Ca^{2+} from the intestine
 - Retention of PO_4^{2-} in the kidney
 - Rickets in children & Osteomalacia in adults

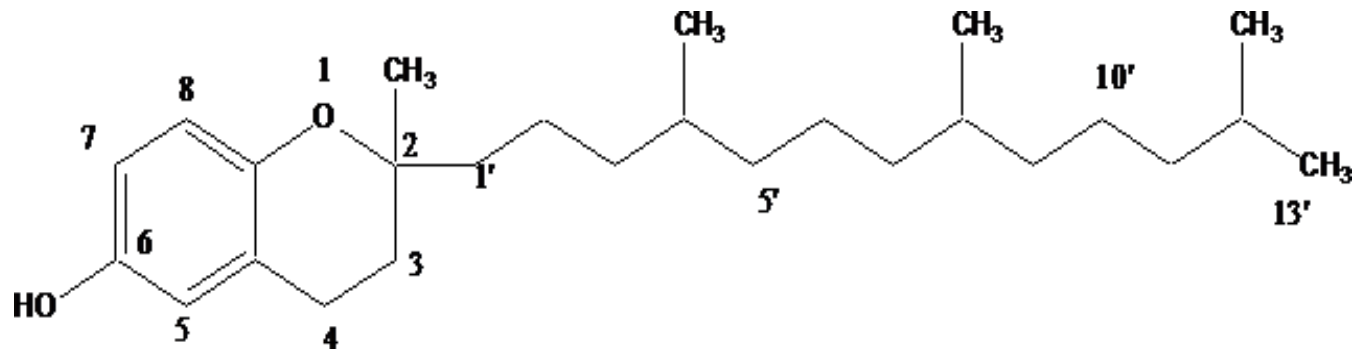
Structural activity relationship vitamin D

- conversion of **C₃- OH** to a **ketone group**
 - **Diminishes activity** greatly but does not completely destroy it
- **Ethers or esters** that **can not be cleaved** in the body have no vitamin D activity
- **Loss or partial degradation** of side chain at **C- 17** or a side chain at C-17 containing **carbonyl/ alcohol groups**
 - Lead to **inactivation** the compound



Vitamin E(tocopherols)

- Have a **chromane ring (Tocol) system**, with an isoprenoid side chain
 - Four tocopherols identified
 - Are **methyl substituted tocol derivatives**
 - – **α , β , γ & δ tocopherol**
 - » α -tocopherol is 5,7,8-trimethyl tocol
 - » β -tocopherol is 5,8-dimethyl tocol
 - » γ -tocopherol is 7,8-dimethyl tocol
 - » δ -tocopherol is 8-methyl tocol

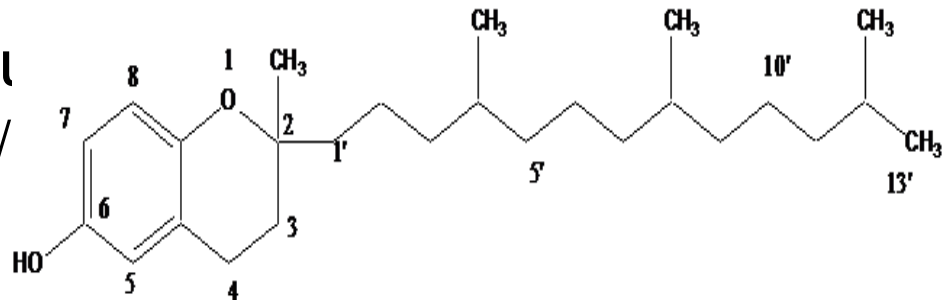


Vitamin E (Tocopherols)

- Stored in adipose tissue
- The most powerful natural antioxidant
- Protect RBC from hemolysis
- Acts as an antimutagen & slows progression of Alzheimer's disease

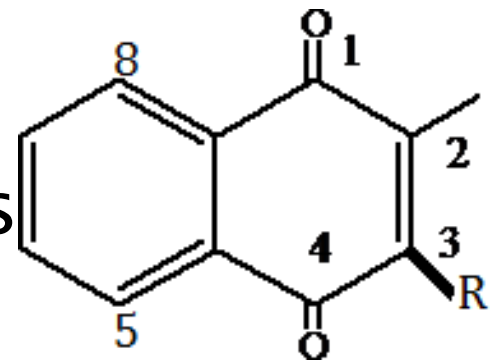
SAR of vitamin E

- In terms of biological activity
 - $\alpha > \beta > \gamma > \delta$
- Esters of tocopherols (acetate, propionate, butyrate)
 - More active than the parent vitamin
- Ethers of tocopherols are inactive
- Activity decreases on
 - Oxidation of tocopherols to respective quinons
 - Replacement of methyl groups with ethyl
 - Reduction in size of the alkyl side chain or introduction of double bonds in the side chain
- Introduction of a 3,4-diol
 - Reduces activity by 2/



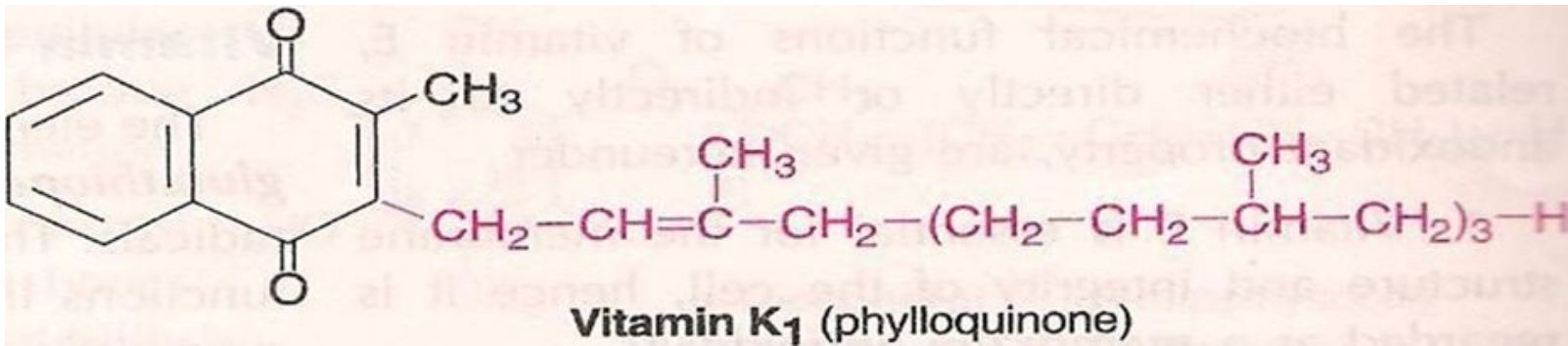
Vitamin K's

- The letter “K” is an abbreviation of the German word “Koagulation Vitamin”
 - Antihemorrhagic factor or coagulation factors
- They are naphthaquinone derivatives with a long isoprenoid side chain. The length of the side chain will differ.
- Importance of Vit K
 - Important in blood clotting process

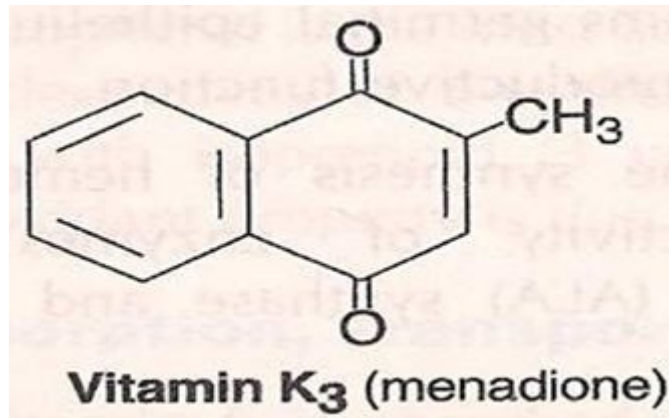
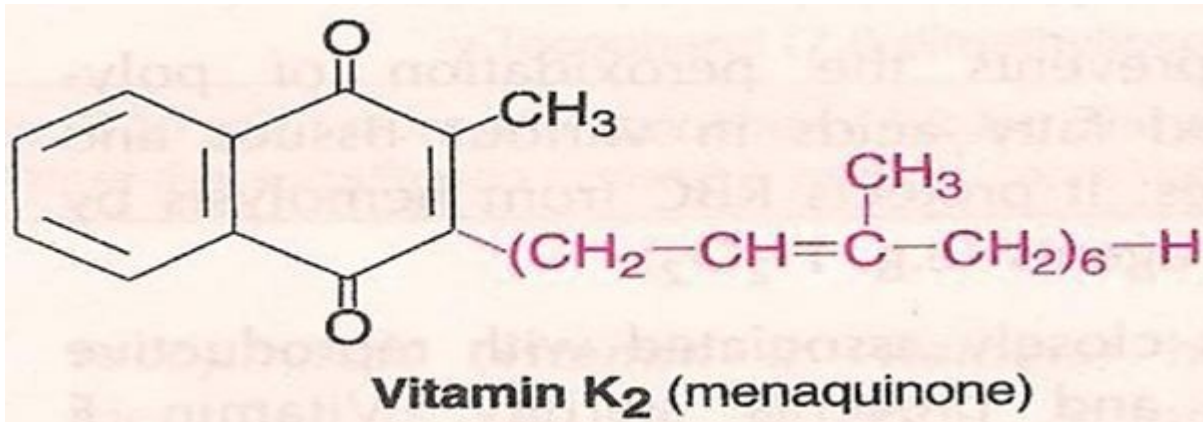


Vitamin K's

- Vitamin K₁
 - Is drug of choice for humans due to
 - Low toxicity
 - Longer duration of action
 - – Compared to menadione & its derivatives
 - Not to be administered to patients receiving anticoagulants
 - Warfarin
 - Coumarins



Other Vitamins K's



Structural activity relationship of Vitamin K's

- Significant biological activity is manifested in compounds with the following structures when:

- Ring A is aromatic

- No substitution on ring A

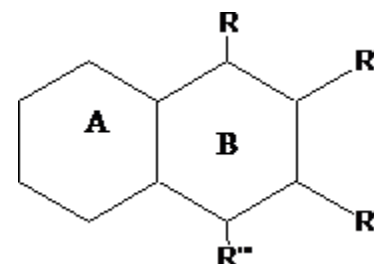
- Ring B is aromatic or heteroaromatic

- R can be OH, O, OR (R = Methyl or Ethyl)

- R' is methyl substituent

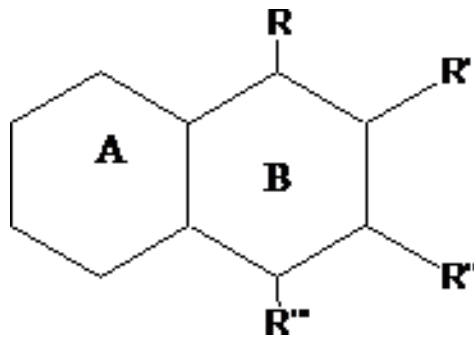
- R'' equals H, SO₃H, N(CH₃)₂ or an alkyl group containing
≥10 carbon atoms

- R''' equals H, OH, NH₂, O, OR (R = Me or Et)



Vitamin K's

- Decreased Anti-hemorrhagic activity obtained when
 - Ring A is substituted
 - R' is an alkyl group larger than Methyl
 - R'' is OH group
 - R''' contains OH in a side chain



Water soluble vitamins

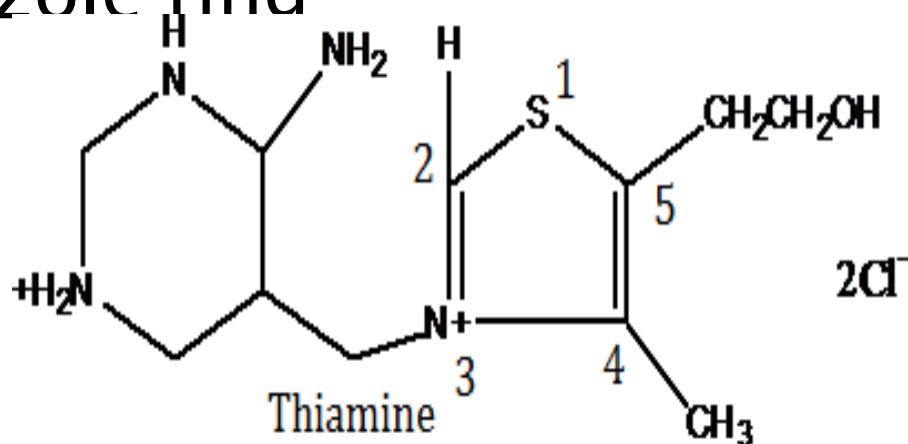
- Water soluble vitamins include
 - The B-complex vitamins
 - Thiamine, Riboflavin, Pyridoxine; Nicotinic acid (Niacin); Pantothenic acid; Biotin; Folic acid; Cyanocobalamine (Vit. B12)
 - Vitamin C (Ascorbic acid)
 - Easily absorbable
- Regular dietary supply is required
- No storage in the body except Vitamin B12
- Act as a co-enzymes

Thiamine (Vitamin B₁)

- Is an antiberiberi agent
- Alcoholism is most common cause of beriberi
- Essential for certain stages in carbohydrate metabolism
- Converted to its active co-enzyme form, thiamine pyrophosphate (TPP), by addition of two phosphate groups

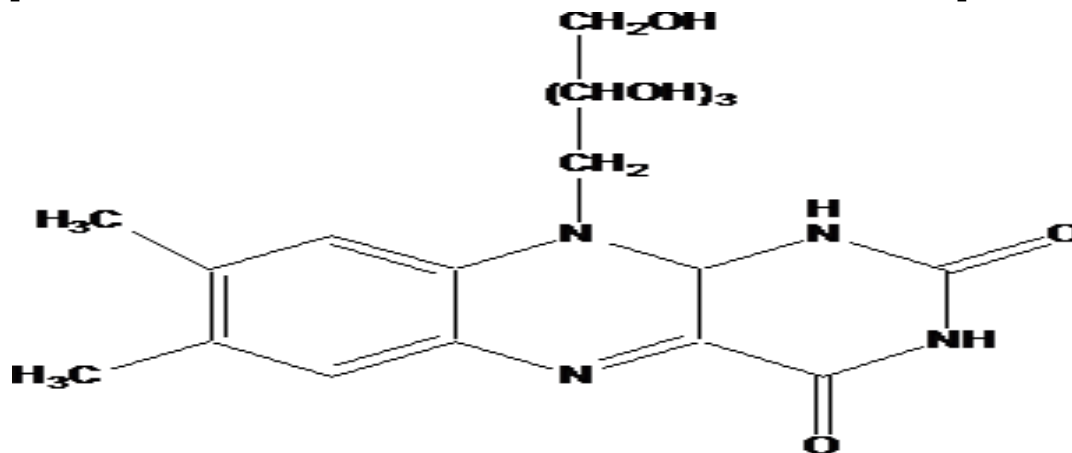
Structural activity relationship of Thiamine

- The following are essential for activity
 - Amino group on pyrimidine ring
 - Hydrogen at position 2 on thiazole ring
 - The 5-hydroxy ethyl group on the thiazole ring



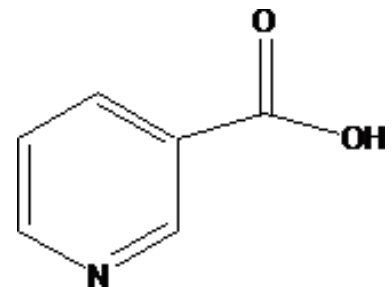
Riboflavin (Vitamin B₂)

- Lactoflavin (milk), Hepatoflavin (Liver), Ovoflavin (Egg) are chemically identical riboflavins
- Contains a dimethyl isoalloxazine ring to which a ribitol is attached (Ribitol is an alcohol of ribose sugar)
- The active forms of riboflavin are:
 - FMN (Flavin Monophosphate Nucleotide) and
 - FAD (Flavin Adenine Dinucleotide)
- Concerned in **carbohydrate metabolism** and necessary for **antibody production, RBC formation, cell respiration and growth**



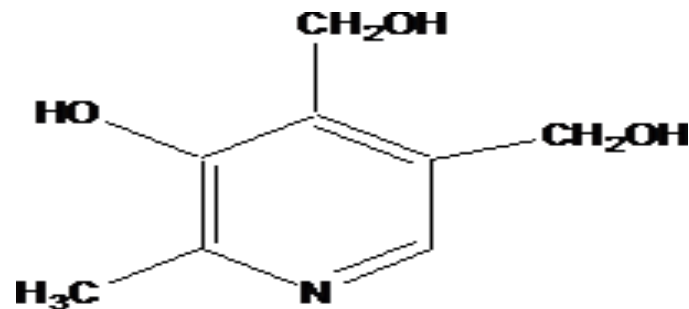
Nicotinic acid (Niacin, Vitamin B₃)

- Also called pellagra preventing factor
- The active form is nicotinamide, present in the tissue
- The active forms, that acts as a coenzyme of nicotinamide are:
 - NAD⁺ (Nicotinamide Adenine Dinucleotide) &
 - NADP⁺ (Nicotinamide Adenine Dinucleotide Phosphate)
- Cause of Niacin deficiency:
 - Malnutrition
 - Dietary deficiency of Tryptophan
 - Lack of Vit B6
- Deficiency of Vit. B₃ causes pellagra (rough skin)
 - Also affects GIT & CNS



Pyridoxine (Vit. B₆)

- Active co-enzyme form is pyridoxal phosphate
- Acts as co-enzyme for many reactions of amino acid metabolism
- Deficiency due to alcoholism
- Peripheral neuritis





Thank you